DEPARTMENT OF TECHNOLOGY

Machine Learning Laboratory

Experiment 03

Multiple Linear Regression

Learning Objective

To implement Multiple Linear Regression for predicting a continuous target variable using multiple input features, and evaluate the performance of the trained model using regression evaluation metrics.

1 Aim

To implement a Multiple Linear Regression model for house price prediction using multiple features and evaluate the model using appropriate regression metrics.

2 Learning Objectives

After completing this experiment, students will be able to:

1. Load and inspect a CSV dataset using Pandas.
2. Identify independent and dependent variables.
3. Perform basic exploratory data analysis.
4. Prepare feature matrix X and target variable y.
5. Split the dataset into training and testing sets.
6. Implement Multiple Linear Regression using Scikit-learn.
7. Interpret model coefficients and intercept.
8. Generate predictions for test data.
9. Evaluate the model using MAE, MSE, RMSE and R^2.
10. Interpret the performance of a regression model.

3 Problem Statement

House prices depend on several factors such as:

- Area of the house
- Number of bedrooms
- Number of bathrooms

- Age of the house
- Distance from metro
- Availability of parking
- Floor number
- Number of balconies

Develop a Multiple Linear Regression model to predict the price of a house using these features.

4 Dataset Description

The dataset used in this experiment is:

ML_Lab_Multiple_Linear_Regression_House_Prices.csv

The dataset contains 100 observations and the following variables:

Feature	Description	Role
Area_sqft	Area of house in square feet	Input
Bedrooms	Number of bedrooms	Input
Bathrooms	Number of bathrooms	Input
Age_years	Age of house in years	Input
Distance_to_Metro_km	Distance from metro in km	Input
Parking	Parking availability (0/1)	Input
Floor	Floor number	Input
Balconies	Number of balconies	Input
Price_Lakhs	House price in lakhs	Target

5 Theory

Multiple Linear Regression is an extension of Simple Linear Regression in which multiple independent variables are used to predict one continuous dependent variable.

The general equation is:

$$\boxed{\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p} \tag{1}$$

where:

- $\hat{y}$ = predicted output
- w_0 = intercept or bias
- $w_1, w_2, \ldots, w_p$ = model coefficients
- $x_1, x_2, \ldots, x_p$ = input features

5.1 Model for the Given Dataset

For this experiment, eight input features are available.

Therefore, the model can be represented as:

$$\begin{aligned}\hat{y} = w_0 &+ w_1(\text{Area}) + w_2(\text{Bedrooms}) + w_3(\text{Bathrooms}) \\ &+ w_4(\text{Age}) + w_5(\text{Distance}) + w_6(\text{Parking}) \\ &+ w_7(\text{Floor}) + w_8(\text{Balconies})\end{aligned} \tag{2}$$

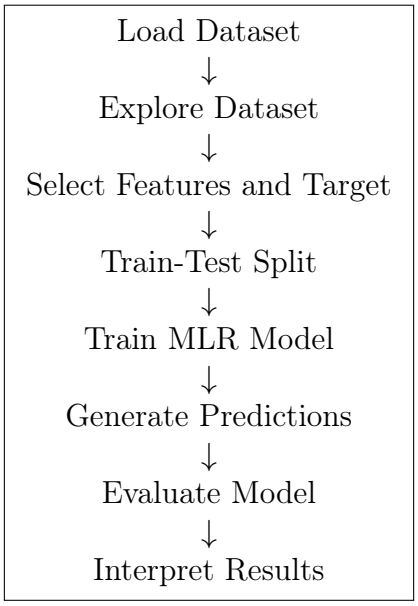
The model learns the values of:

$$w_0, w_1, w_2, \ldots, w_8$$

from the training data.

6 Regression Workflow

The experiment follows the following machine learning workflow:



7 Implementation

7.1 Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
```

7.2 Step 2: Load the Dataset

Load the CSV file using Pandas.

```
df = pd.read_csv(
    "ML_Lab_Multiple_Linear_Regression_House_Prices.csv"
)

print(df.head())
```

7.3 Step 3: Explore the Dataset

Display the basic information about the dataset.

```
print(df.shape)

print(df.info())

print(df.describe())

print(df.isnull().sum())
```

Students should determine:

- Number of observations
- Number of features
- Data types
- Presence of missing values
- Basic statistical properties

7.4 Step 4: Separate Features and Target

The target variable is:

`Price_Lakhs`

The remaining selected variables are used as input features.

```
features = [
    'Area_sqft',
    'Bedrooms',
    'Bathrooms',
    'Age_years',
    'Distance_to_Metro_km',
    'Parking',
    'Floor',
    'Balconies'
]

X = df[features]

y = df['Price_Lakhs']
```

Therefore:

$$\boxed{X = \text{Feature Matrix}}$$

and

$$\boxed{y = \text{Target Variable}}$$

7.5 Step 5: Split the Dataset

Divide the dataset into training and testing subsets.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

In this experiment:

$$80\% \rightarrow \text{Training Data}$$

$$20\% \rightarrow \text{Testing Data}$$

The training data is used to learn the model, while the testing data is used to evaluate its performance on unseen observations.

7.6 Step 6: Create the Model

Create an instance of the Linear Regression model.

```
model = LinearRegression()
```

7.7 Step 7: Train the Model

Train the model using the training data.

```
model.fit(X_train, y_train)
```

The `fit()` method estimates the coefficients and intercept from the training data.

7.8 Step 8: Display Model Parameters

Display the intercept and coefficients.

```
print("Intercept:", model.intercept_)

for feature, coefficient in zip(
       features,
       model.coef_):

    print(feature, ":", coefficient)
```

The resulting model has the form:

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \cdots + w_8x_8 \tag{3}$$

The values of the coefficients are learned from the training data.

7.9 Step 9: Generate Predictions

Use the trained model to predict prices for the test dataset.

```
y_pred = model.predict(X_test)
```

Create a comparison between actual and predicted values.

```
comparison = pd.DataFrame({
    'Actual Price': y_test.values,
    'Predicted Price': y_pred
})

print(comparison)
```

8 Model Evaluation

The trained model must be evaluated using regression evaluation metrics.

The following metrics are used:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

8.1 Mean Absolute Error

$$\boxed{MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|} \tag{4}$$

MAE represents the average absolute difference between actual and predicted values.

8.2 Mean Squared Error

$$\boxed{MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{5}$$

MSE gives greater importance to larger errors because the errors are squared.

8.3 Root Mean Squared Error

$$\boxed{RMSE = \sqrt{MSE}} \tag{6}$$

RMSE is expressed in the same units as the target variable.

8.4 Coefficient of Determination

$$\boxed{R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}} \tag{7}$$

where:

$$\bar{y} = \text{mean of the actual target values}$$

A higher R^2 generally indicates that the model explains a greater proportion of the variation in the target variable.

8.5 Calculate Evaluation Metrics

```
mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, y_pred)

print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R2 :", r2)
```

9 Actual vs Predicted Values

Create a scatter plot to compare actual and predicted house prices.

```
plt.scatter(y_test, y_pred)

plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")

plt.title(
    "Actual vs Predicted House Prices"
)

plt.show()
```

An ideal model would produce predictions close to:

$$\boxed{\hat{y} = y} \tag{8}$$

10 Prediction for a New House

Consider a new house with the following characteristics:

Feature	Value
Area	1800 sq.ft
Bedrooms	3
Bathrooms	2
Age	5 years
Distance to Metro	3.5 km
Parking	1
Floor	5
Balconies	2

Create the input using the same feature order used during training.

```
new_house = [[
    1800,
    3,
    2,
    5,
    3.5,
    1,
    5,
    2
]]

predicted_price = model.predict(new_house)

print(
    "Predicted House Price:",
    predicted_price[0],
    "Lakhs"
)
```

11 Observations

Record the following values after executing the program:

Measure	Value
Number of observations	
Number of features	
Training observations	
Testing observations	
Intercept	
MAE	
MSE	
RMSE	
R^2 Score	
Predicted price of new house	

12 Result

The Multiple Linear Regression model was successfully implemented to predict house prices using multiple input features.

The model was evaluated using MAE, MSE, RMSE and R^2 score.

13 Post-Lab Questions

1. What is the difference between Simple Linear Regression and Multiple Linear Regression?
2. Why is the target variable separated from the feature matrix?
3. What is the purpose of `train_test_split()`?
4. Why should the test data not be used for training the model?
5. What does the `fit()` method do?
6. What does the `predict()` method do?
7. What does each coefficient in a Multiple Linear Regression model represent?
8. What is the role of the intercept?
9. Why can MSE be larger than MAE?
10. What is the difference between MSE and RMSE?
11. What does R^2 indicate?
12. If the R^2 score is low, what might be possible reasons?
13. What would happen if an irrelevant feature were added to the model?
14. Why should the same feature order be maintained when predicting a new observation?

14 Additional Tasks

1. Train the model using only:

   $$\text{Area_sqft}, \text{Bedrooms}, \text{Bathrooms}$$

   Compare its performance with the model using all eight features.
2. Plot actual versus predicted prices for both models.
3. Compare their MAE, RMSE and R^2 values.
4. Determine whether adding more features improved the model performance.

15 Important Note

Important

In this experiment, students are only required to implement and evaluate Multiple Linear Regression.
Feature engineering, feature selection, feature scaling, and regularization will be covered in subsequent experiments.

Key Learning Flow

Dataset → Features and Target
→ Train-Test Split
→ MLR Model
→ Training
→ Prediction
→ Evaluation